



Heaven's Light is Our Guide

Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh

Course Code: CSE 2103
Course Title: Digital Logic Design

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Common Number Systems

System	Base	Symbols	Used by humans?	Used in computers?
Decimal	10	0, 1, ... 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, ... 7	No	No
Hexa-decimal	16	0, 1, ... 9, A, B, ... F	No	No

Quantities/Counting (1 of 3)

Decimal	Binary	Octal	Hexa- decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7

Quantities/Counting (2 of 3)

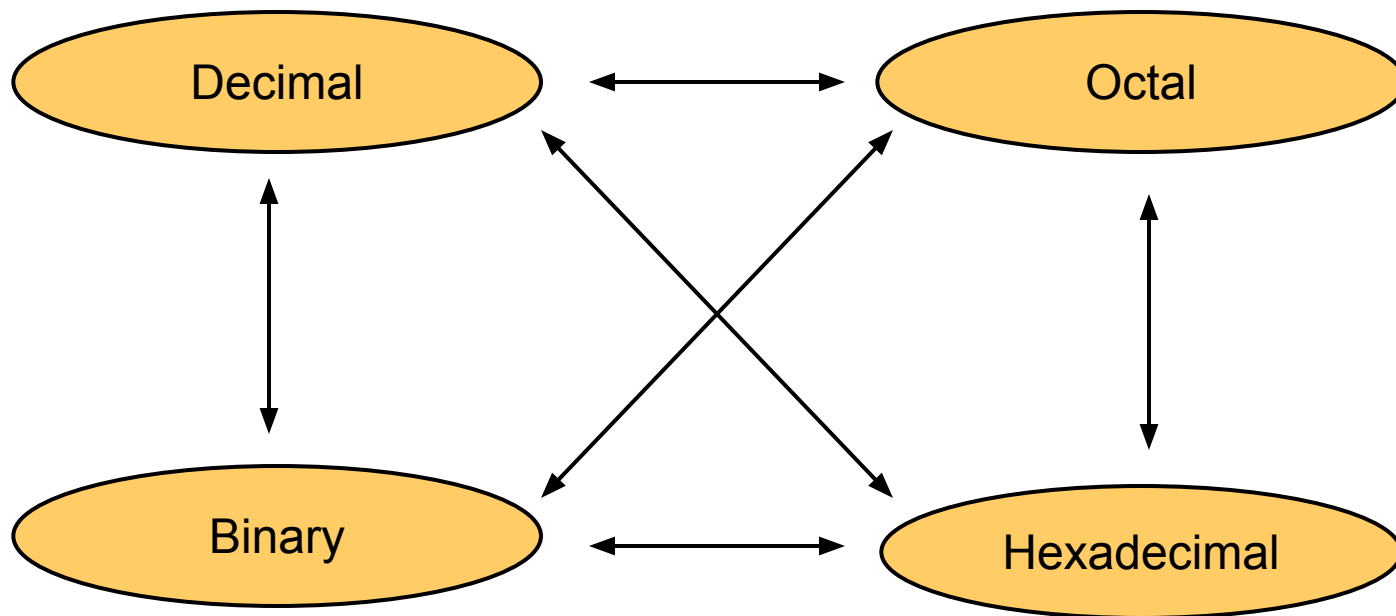
Decimal	Binary	Octal	Hexa- decimal
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Quantities/Counting (3 of 3)

Decimal	Binary	Octal	Hexa- decimal
16	10000	20	10
17	10001	21	11
18	10010	22	12
19	10011	23	13
20	10100	24	14
21	10101	25	15
22	10110	26	16
23	10111	27	17

Conversion Among Bases

The possibilities:

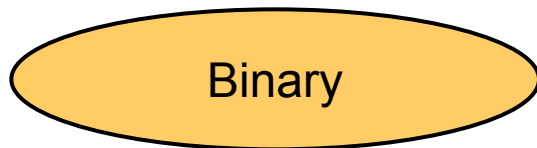
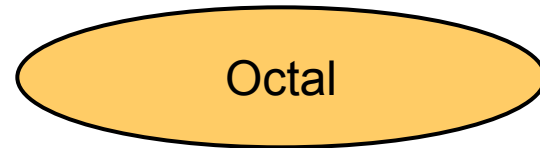
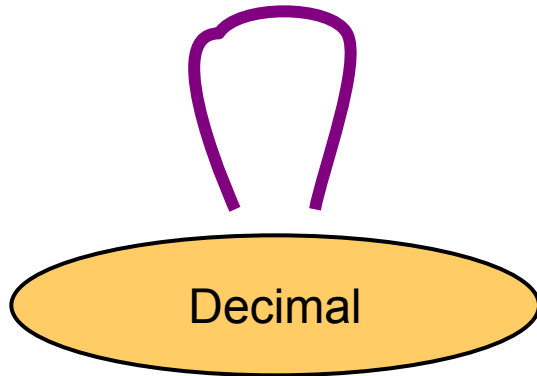


Quick Example

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$

Base

Decimal to Decimal (just for fun)



Next
slide...

Weight

$$125_{10} \Rightarrow 5 \times 10^0 = 5$$

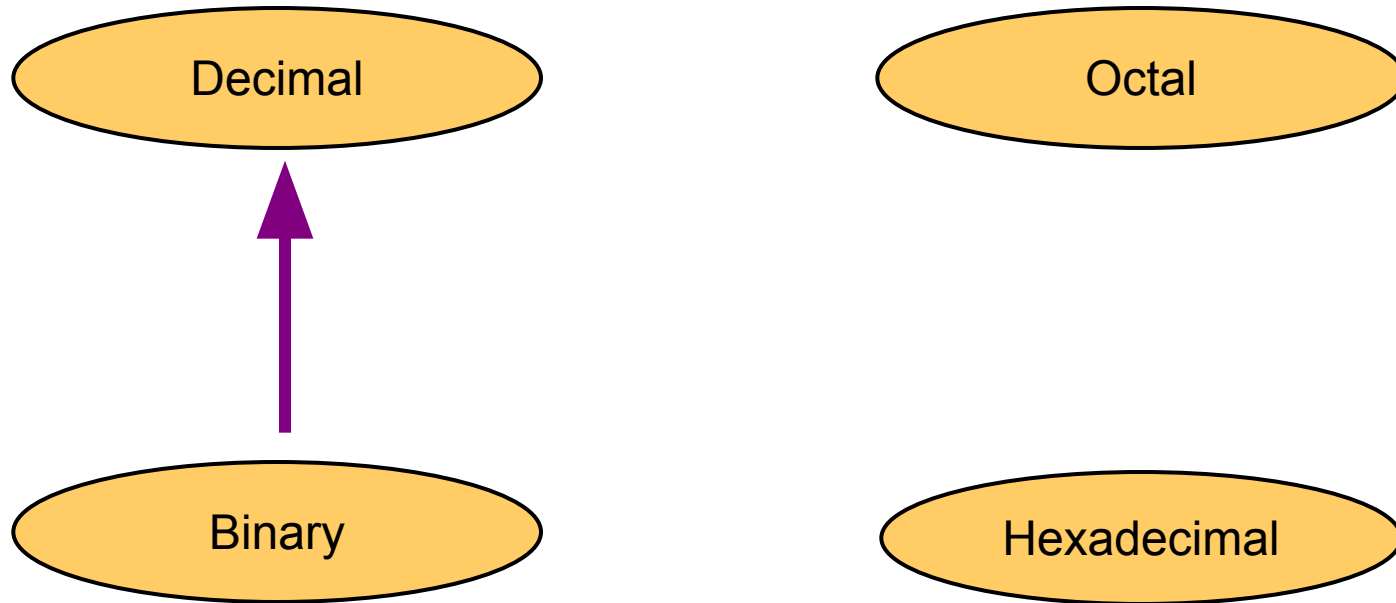
$$2 \times 10^1 = 20$$

$$1 \times 10^2 = 100$$

125

Base

Binary to Decimal



Binary to Decimal

Technique

- Multiply each bit by 2^n , where n is the “weight” of the bit.
- The weight is the position of the bit, starting from 0 on the right
- Add the results

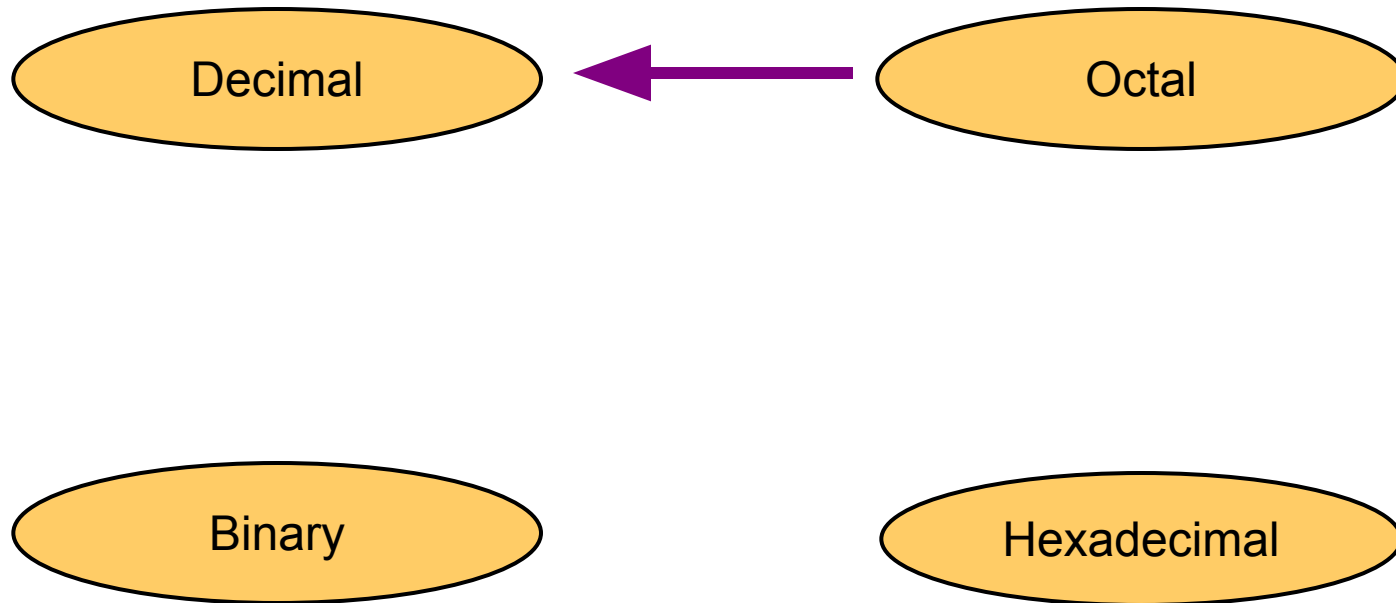
Example

Bit "0"

$101011_2 \Rightarrow$

1	x	2^0	=	1
1	x	2^1	=	2
0	x	2^2	=	0
1	x	2^3	=	8
0	x	2^4	=	0
1	x	2^5	=	32
				43_{10}

Octal to Decimal



Octal to Decimal

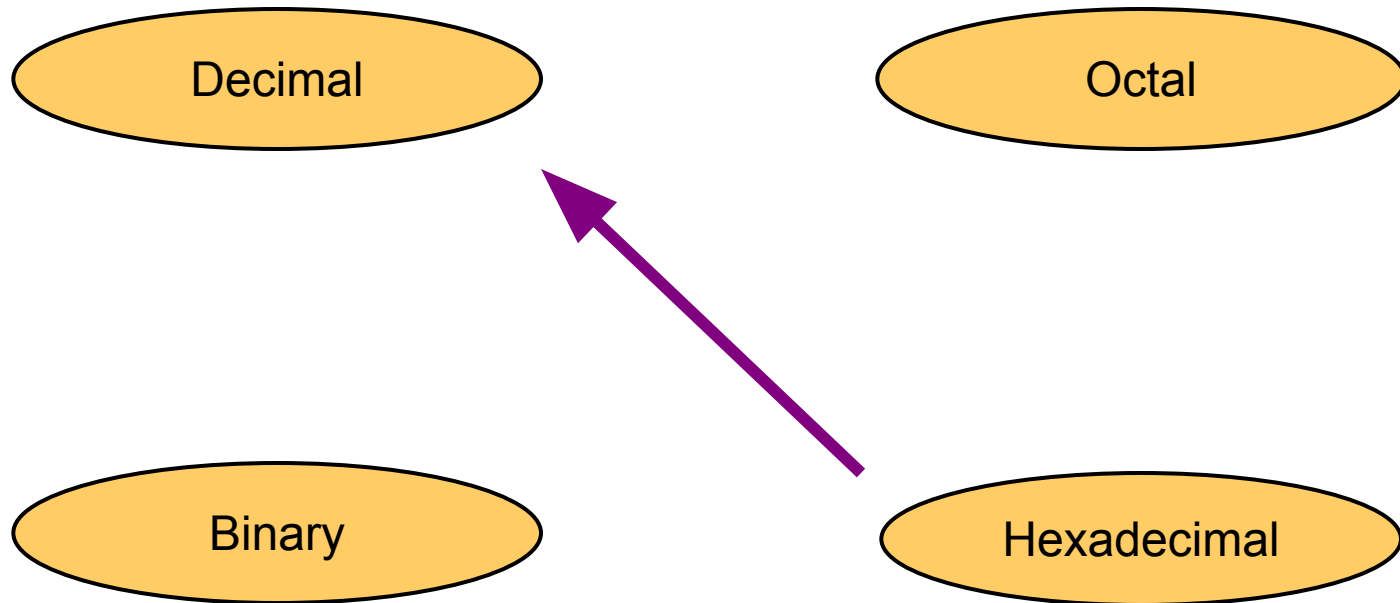
Technique

- Multiply each bit by 8^n , where n is the “weight” of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results

Example

$$\begin{array}{rcll} 724_8 & \Rightarrow & 4 \times 8^0 & = & 4 \\ & & 2 \times 8^1 & = & 16 \\ & & 7 \times 8^2 & = & 448 \\ & & & & \hline & & & & 468_{10} \end{array}$$

Hexadecimal to Decimal



Hexadecimal to Decimal

Technique

- Multiply each bit by 16^n , where n is the “weight” of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results

Example

$$\begin{array}{rcll} \text{ABC}_{16} \Rightarrow & \text{C} \times 16^0 & = 12 \times 1 & = 12 \\ & \text{B} \times 16^1 & = 11 \times 16 & = 176 \\ & \text{A} \times 16^2 & = 10 \times 256 & = 2560 \\ & & & \hline & & & 2748_{10} \end{array}$$

Decimal to Binary

Decimal

Octal



Binary

Hexadecimal

Decimal to Binary

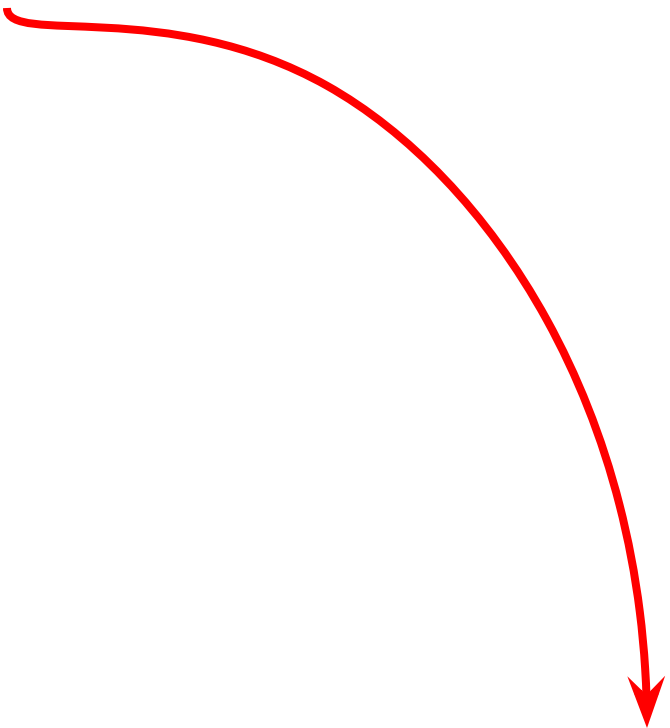
Technique

- Divide by two, keep track of the remainder
- First remainder is bit 0 (LSB, least-significant bit)
- Second remainder is bit 1
- Etc.

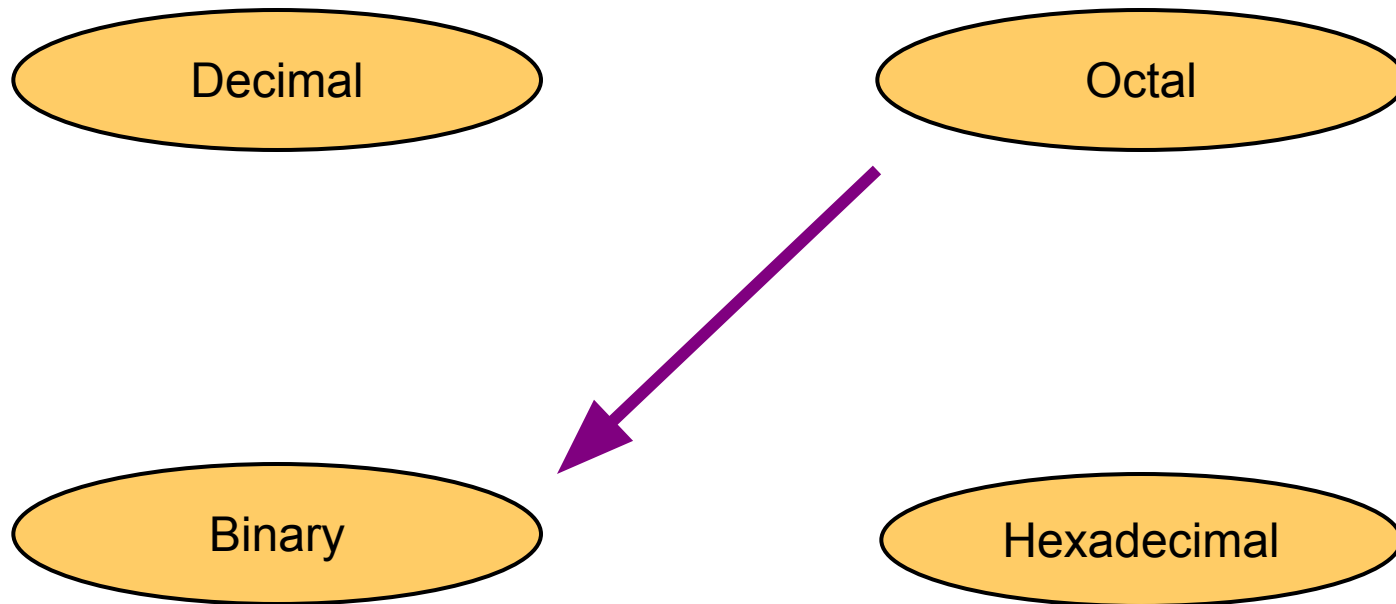
Example

$$125_{10} = ?_2$$

2		125	
2		62	1
2		31	0
2		15	1
2		7	1
2		3	1
2		1	1
		0	1


$$125_{10} = 1111101_2$$

Octal to Binary



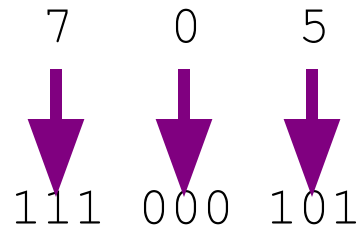
Octal to Binary

Technique

Convert each octal digit to a 3-bit equivalent binary representation

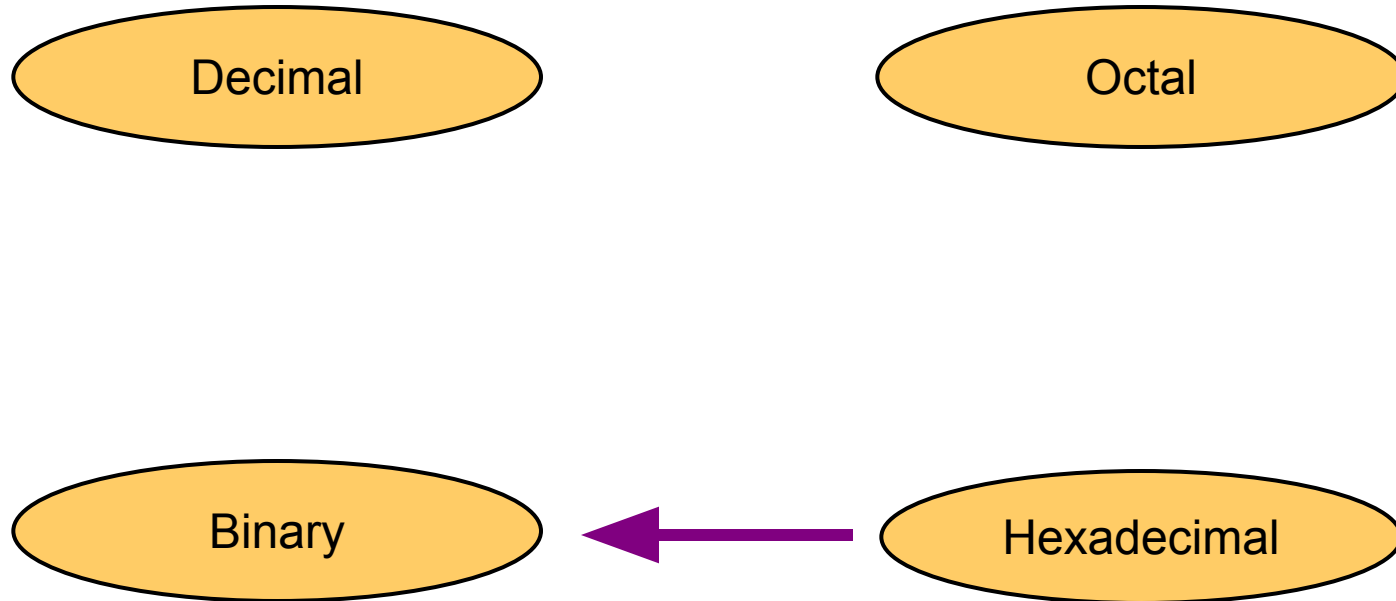
Example

$$705_8 = ?_2$$



$$705_8 = 111000101_2$$

Hexadecimal to Binary



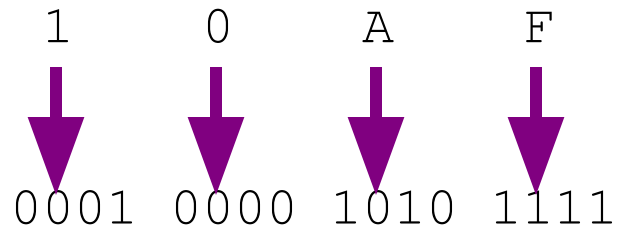
Hexadecimal to Binary

Technique

Convert each hexadecimal digit to a 4-bit equivalent binary representation

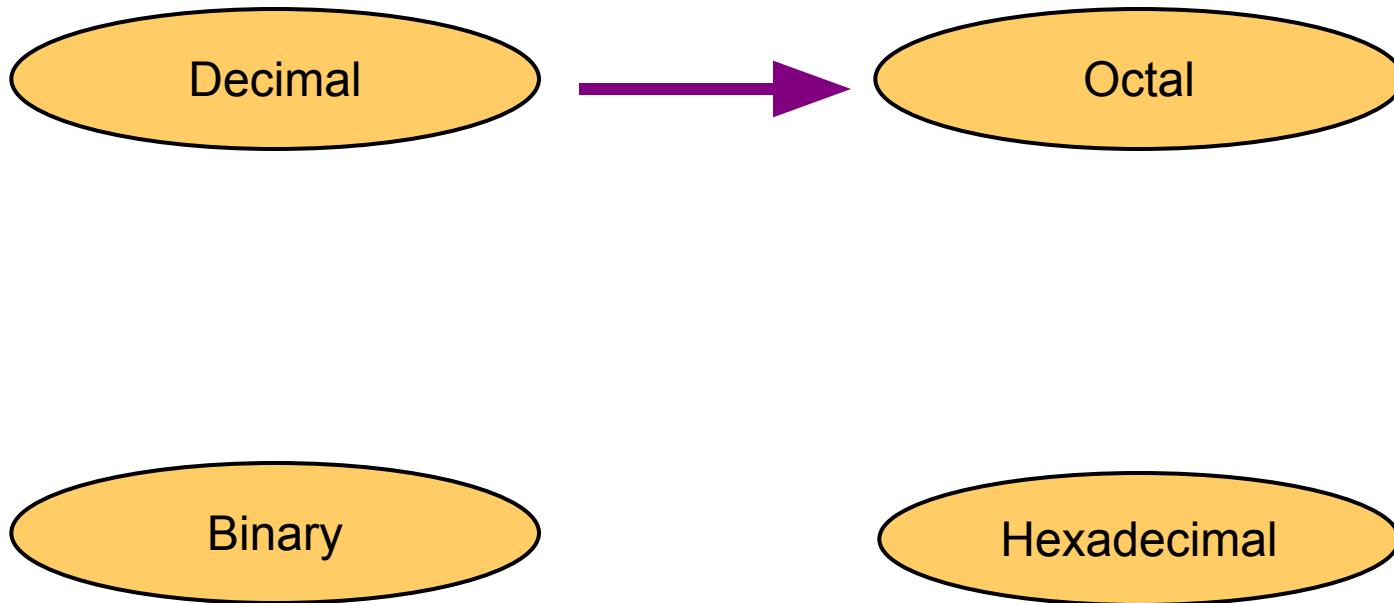
Example

$$10AF_{16} = ?_2$$



$$10AF_{16} = 0001000010101111_2$$

Decimal to Octal



Decimal to Octal

Technique

- Divide by 8
- Keep track of the remainder

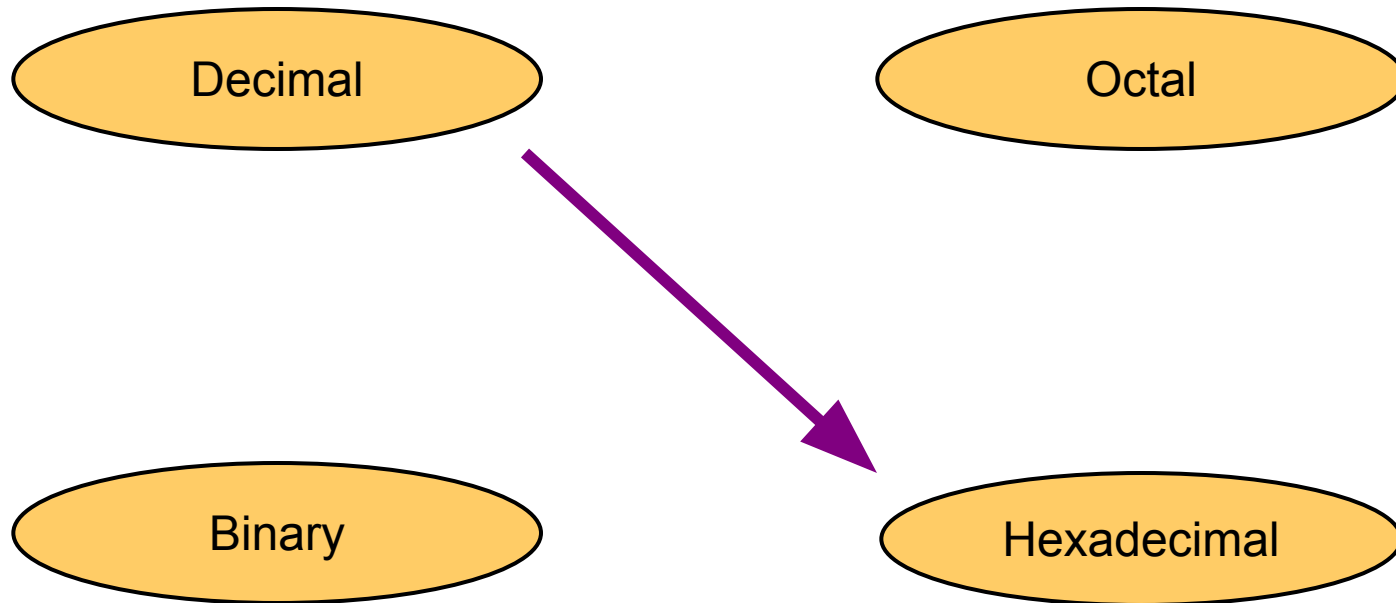
Example

$$1234_{10} = ?_8$$

8		1234	
8		154	2
8		19	2
8		2	3
		0	2

$$1234_{10} = 2322_8$$

Decimal to Hexadecimal



Decimal to Hexadecimal

Technique

- Divide by 16
- Keep track of the remainder

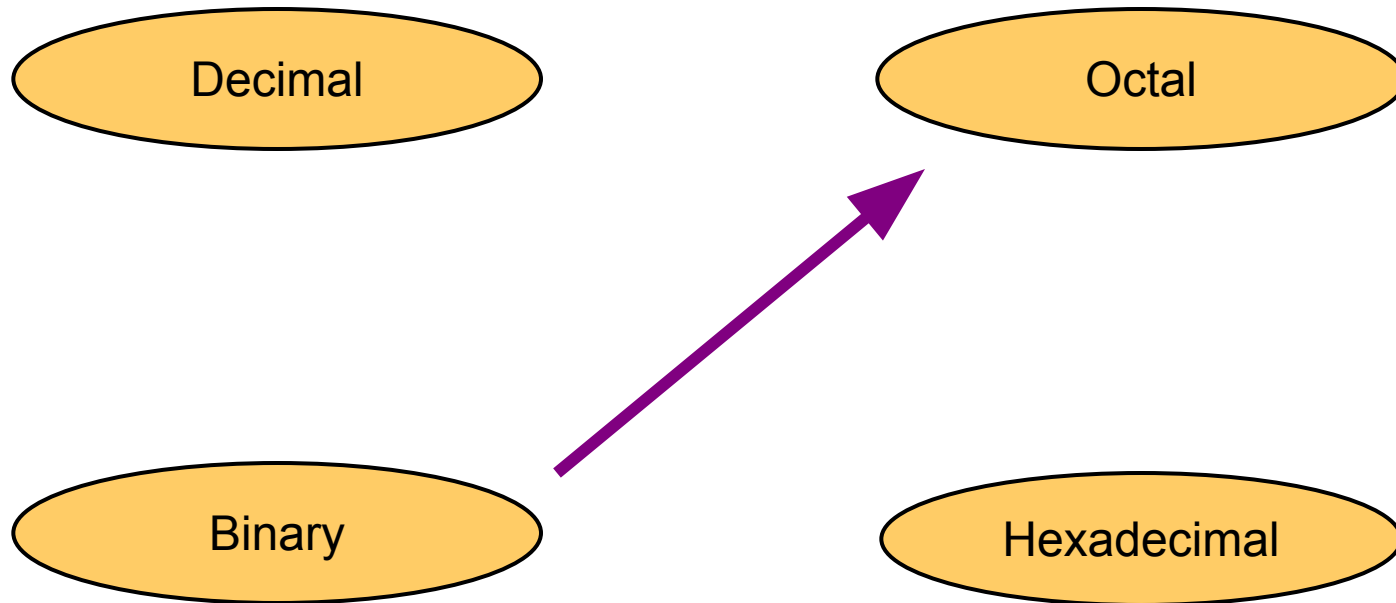
Example

$$1234_{10} = ?_{16}$$

16	1234	
16	77	2
16	4	13 = D
	0	4

$$1234_{10} = 4D2_{16}$$

Binary to Octal



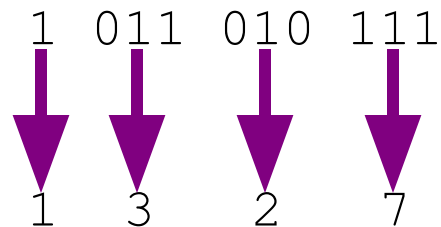
Binary to Octal

Technique

- Group bits in threes, starting on right
- Convert to octal digits

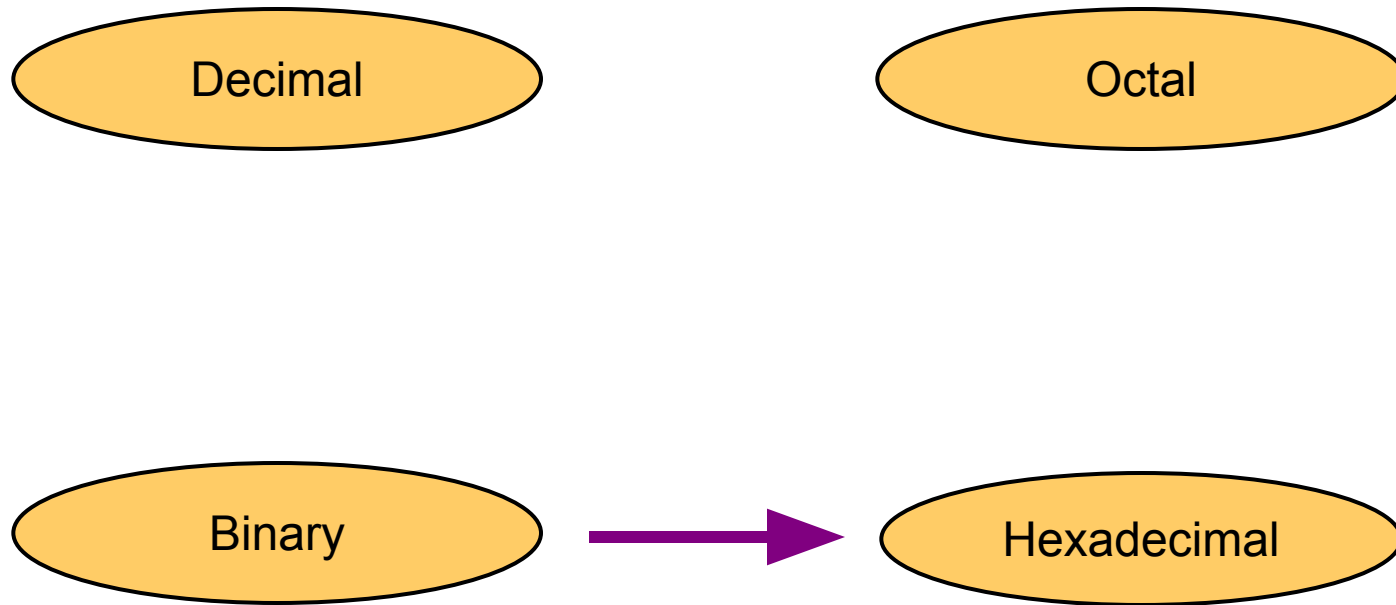
Example

$$1011010111_2 = ?_8$$



$$1011010111_2 = 1327_8$$

Binary to Hexadecimal



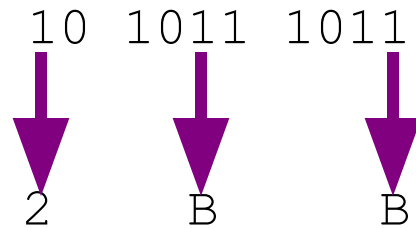
Binary to Hexadecimal

Technique

- Group bits in fours, starting on right
- Convert to hexadecimal digits

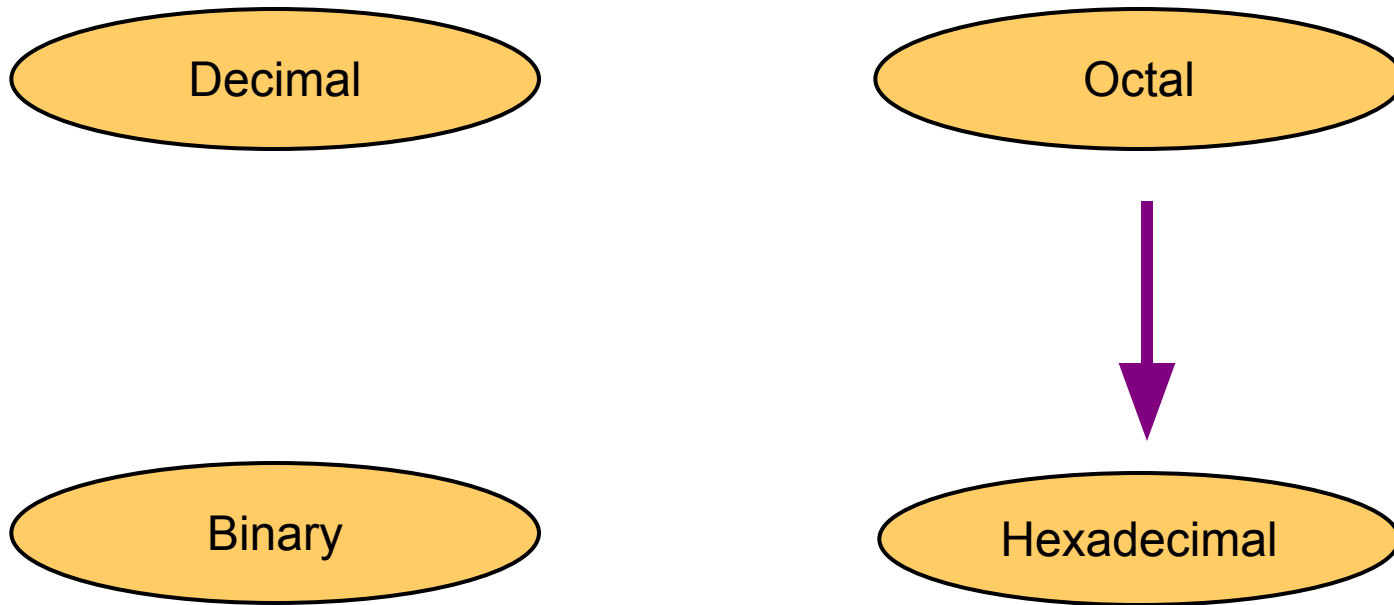
Example

$$1010111011_2 = ?_{16}$$



$$1010111011_2 = 2BB_{16}$$

Octal to Hexadecimal



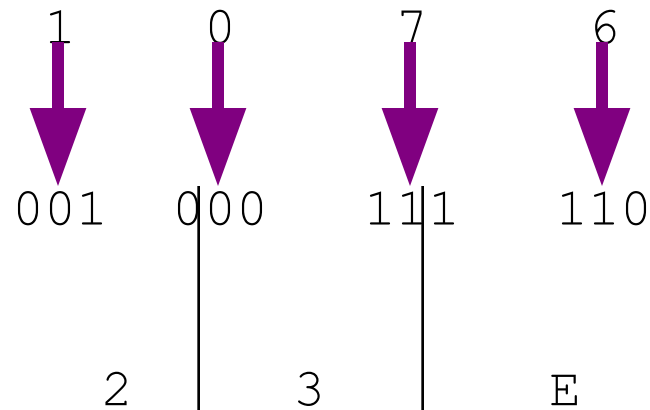
Octal to Hexadecimal

Technique

- Use binary as an intermediary

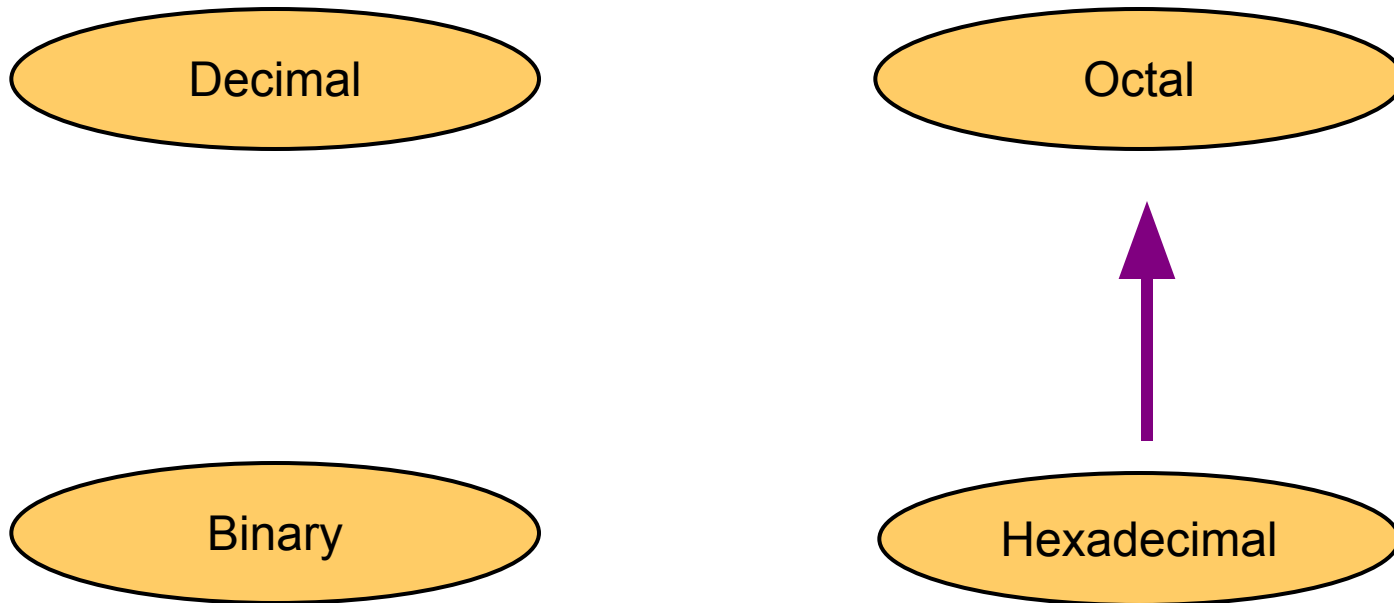
Example

$$1076_8 = ?_{16}$$



$$1076_8 = 23E_{16}$$

Hexadecimal to Octal



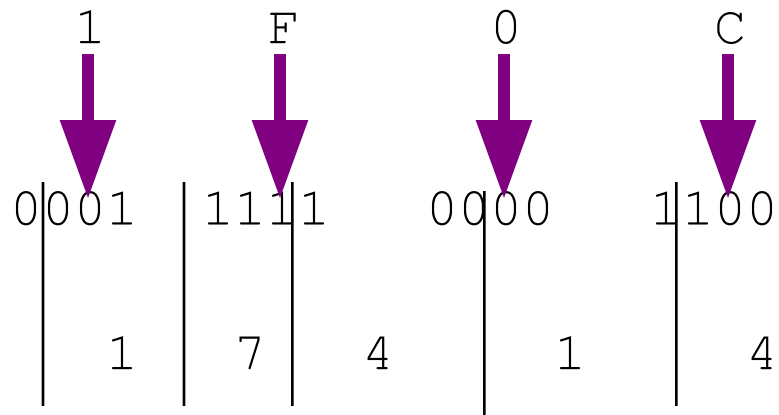
Hexadecimal to Octal

Technique

- Use binary as an intermediary

Example

$$1F0C_{16} = ?_8$$



$$1F0C_{16} = 17414_8$$

Exercise – Convert ...

Decimal	Binary	Octal	Hexa- decimal
33			
	1110101		
		703	
			1AF

Don't use a calculator!

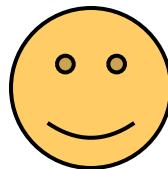
Skip answer

Answer

Exercise – Convert ...

Answer

Decimal	Binary	Octal	Hexa- decimal
33	100001	41	21
117	1110101	165	75
451	111000011	703	1C3
431	110101111	657	1AF



Common Powers (1 of 2)

Base 10

Power	Preface	Symbol	Value
10^{-12}	pico	p	.000000000001
10^{-9}	nano	n	.000000001
10^{-6}	micro	μ	.000001
10^{-3}	milli	m	.001
10^3	kilo	k	1000
10^6	mega	M	1000000
10^9	giga	G	1000000000
10^{12}	tera	T	1000000000000

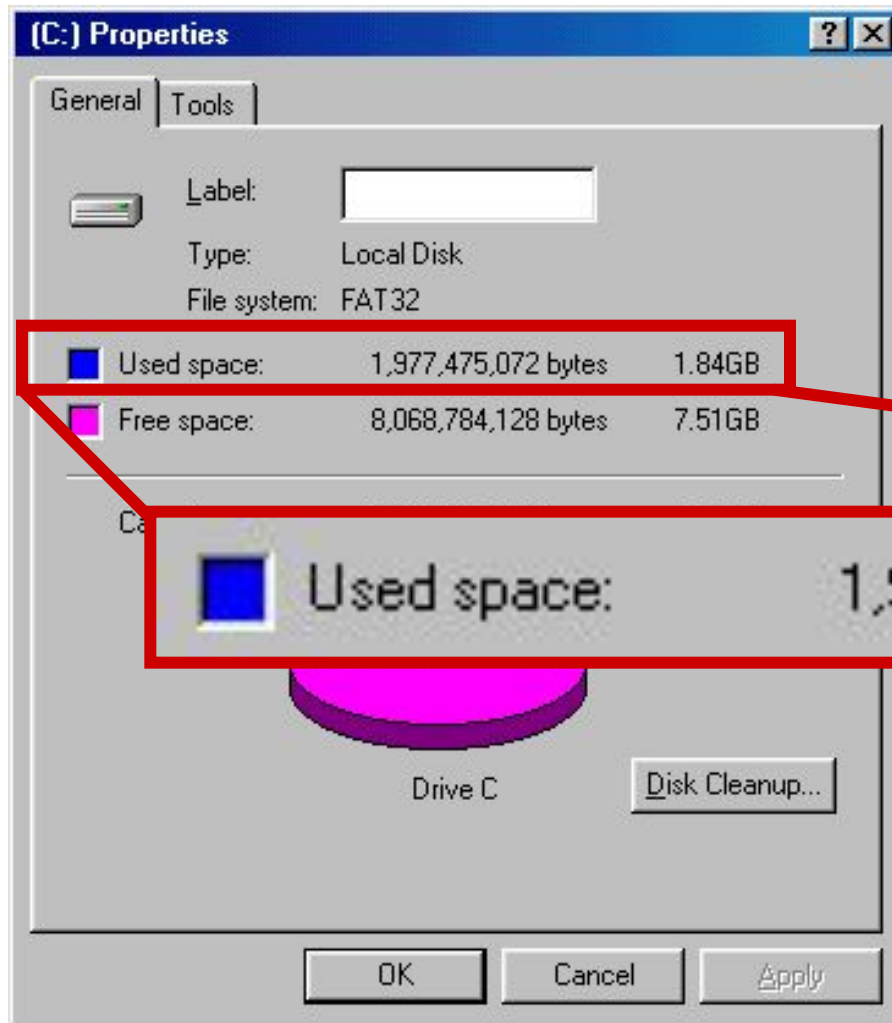
Common Powers (2 of 2)

Base 2

Power	Preface	Symbol	Value
2^{10}	kilo	k	1024
2^{20}	mega	M	1048576
2^{30}	Giga	G	1073741824

- What is the value of “k”, “M”, and “G”?
- In computing, particularly w.r.t. memory, the base-2 interpretation generally applies

Example



In the lab...

1. Double click on My Computer
2. Right click on C:
3. Click on Properties

$$/ 2^{30} =$$

Exercise – Free Space

Determine the “free space” on all drives on a machine in the lab

Drive	Free space	
	Bytes	GB
A:		
C:		
D:		
E:		
etc.		

Review – multiplying powers

For common bases, add powers

$$a^b \times a^c = a^{b+c}$$

$$2^6 \times 2^{10} = 2^{16} = 65,536$$

or...

$$2^6 \times 2^{10} = 64 \times 2^{10} = 64k$$

Binary Addition (1 of 2)

Two 1-bit values

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	10



Binary Addition (2 of 2)

Two n -bit values

Add individual bits

Propagate carries

E.g.,

$$\begin{array}{r} \\ 10101 \\ + 11001 \\ \hline 101110 \end{array} \qquad \begin{array}{r} 21 \\ + 25 \\ \hline 46 \end{array}$$

Multiplication (1 of 3)

Decimal (just for fun)

$$\begin{array}{r} 35 \\ \times 105 \\ \hline 175 \\ 000 \\ 35 \\ \hline 3675 \end{array}$$

Multiplication (2 of 3)

Binary, two 1-bit values

A	B	$A \times B$
0	0	0
0	1	0
1	0	0
1	1	1

Multiplication (3 of 3)

Binary, two n -bit values

As with decimal values

E.g.,

$$\begin{array}{r} 1110 \\ \times 1011 \\ \hline 1110 \\ 1110 \\ 0000 \\ 1110 \\ \hline 10011010 \end{array}$$

Fractions

Decimal to decimal (just for fun)

$$\begin{array}{rcl} 3.14 & \Rightarrow & 4 \times 10^{-2} = 0.04 \\ & & 1 \times 10^{-1} = 0.1 \\ & & 3 \times 10^0 = 3 \end{array} \quad \underline{\hspace{1cm}}$$

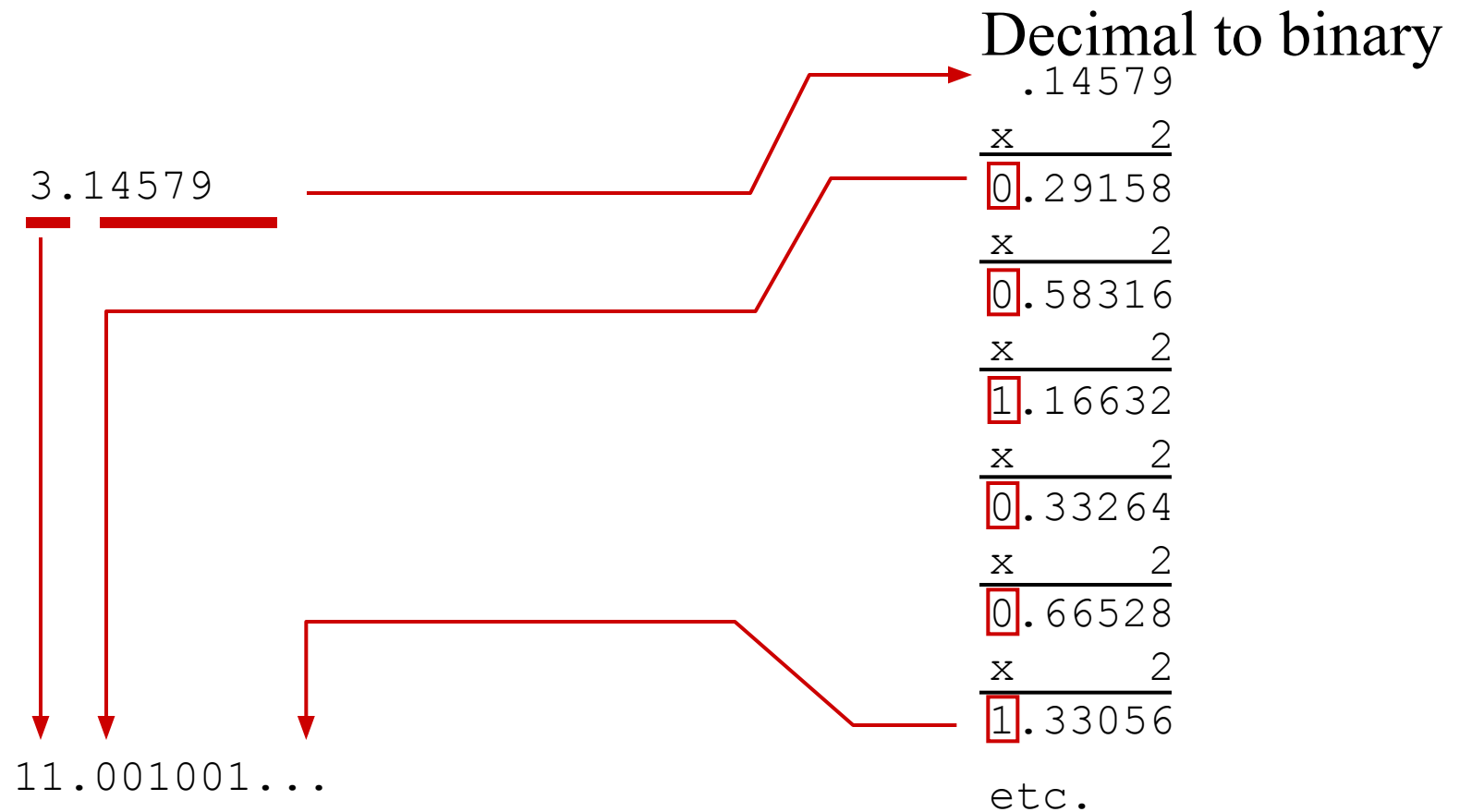
3.14

Fractions

Binary to decimal

$$\begin{array}{rcll} 10.1011 & => & 1 \times 2^{-4} & = 0.0625 \\ & & 1 \times 2^{-3} & = 0.125 \\ & & 0 \times 2^{-2} & = 0.0 \\ & & 1 \times 2^{-1} & = 0.5 \\ & & 0 \times 2^0 & = 0.0 \\ & & 1 \times 2^1 & = 2.0 \\ & & & \hline & & & 2.6875 \end{array}$$

Fractions



Exercise – Convert ...

Decimal	Binary	Octal	Hexa- decimal
29.8			
	101.1101		
		3.07	
			C.82

Don't use a calculator!

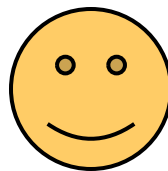
Skip answer

Answer

Exercise – Convert ...

Answer

Decimal	Binary	Octal	Hexa- decimal
29.8	11101.110011...	35.63...	1D.CC...
5.8125	101.1101	5.64	5.D
3.109375	11.000111	3.07	3.1C
12.5078125	1100.10000010	14.404	C.82



Thank
You



Next topic